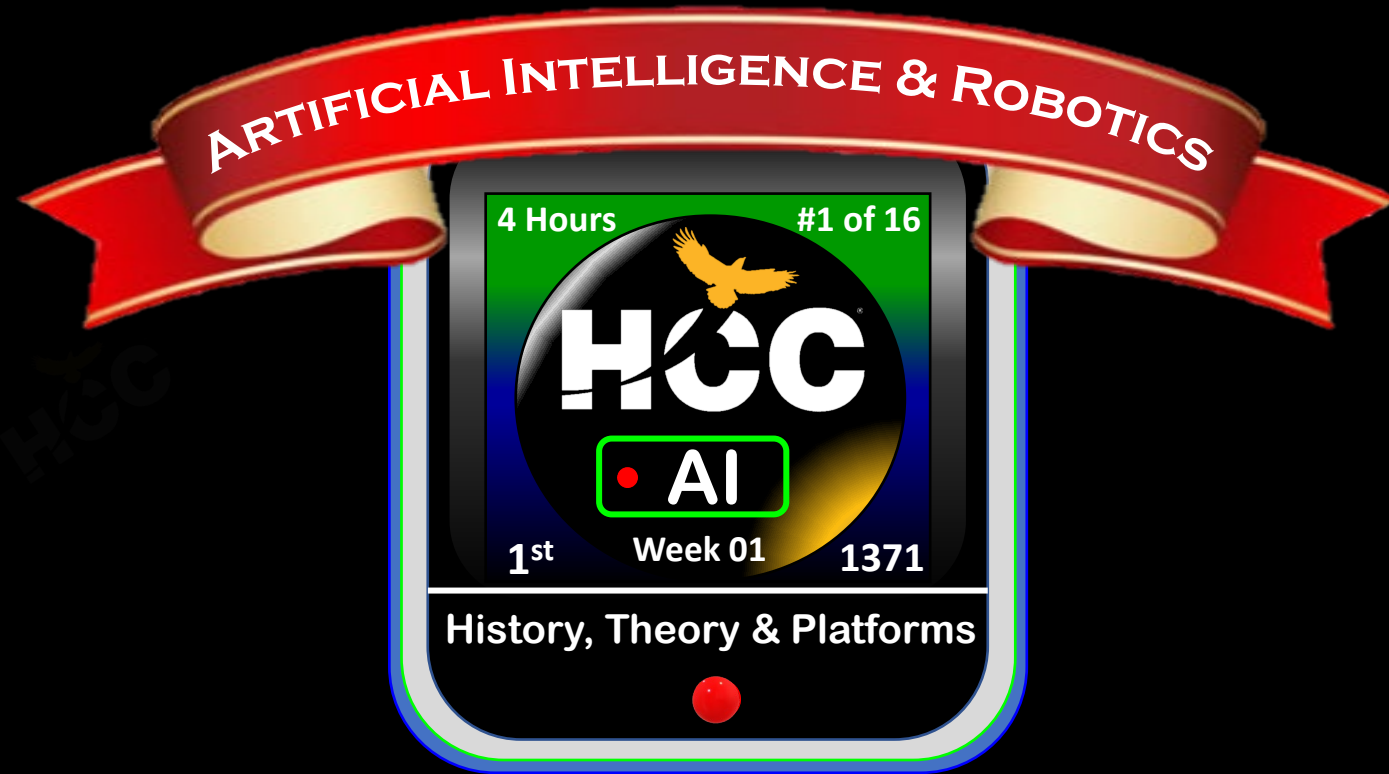


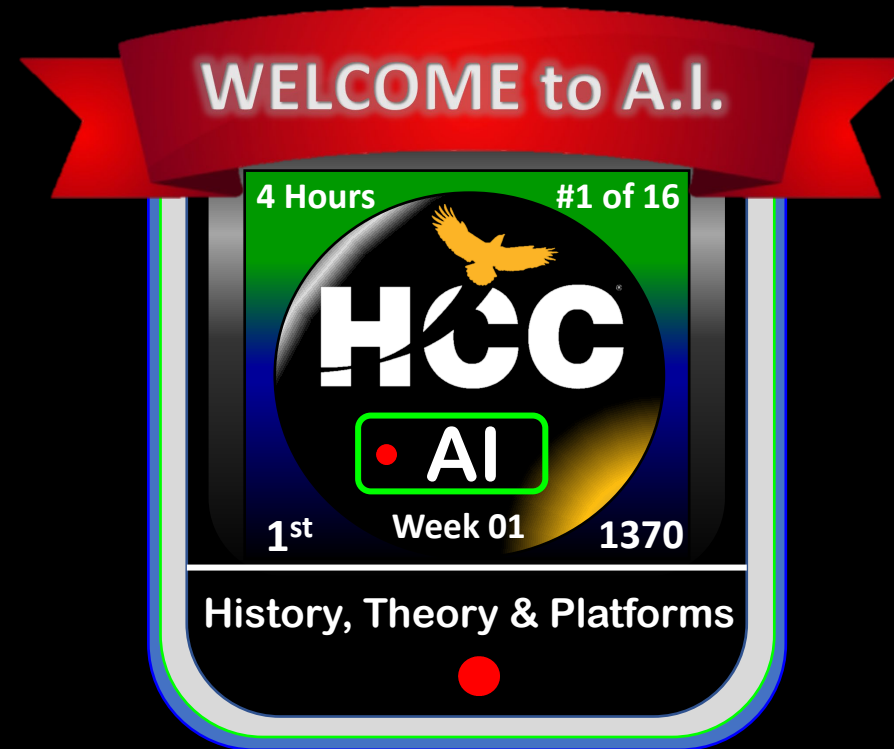
Welcome To The New Bachelor's Program



We are delighted to have you here

©

Artificial Intelligence Course 1



Introduction to A.I. Module 1

©

History, Theory and Platforms Overview



Introduction to A.I. – Course Setup

Online - LIVE

2 of 70

96.00

A.I. 1370



Welcome Course Note

Why is the Artificial Intelligence Course of Study being offered by Houston Community College so Pivotal? **Access, Cost and Reach**

The A.I. Course is a first in several categories:

1. HCC is one of the first Community Colleges in the Country to Offer an AAS in A.I. and a Bachelor's Degree in Artificial Intelligence and Robotics – World Class
2. The Program offers the finest material, the finest subject matter experts and State-of-the-Art methods available to teach “Best Practices” of A.I.
3. The course integrates a thorough use of “The Best” Fundamentals from companies Like AWS, Intel, Nvidia and Open Courseware – State of the Art
4. The Program immerses the students into the “Real World Applications” and Issues like Bias, Ethics, Cyber, Autonomy, CV, ML, NLP in a “Hand's-On” manner
5. **The Program is using A.I. to teach A.I. by embedding “Smart Design Templates” to insure scalability, flexibility, quality, package-ability and assessment/metrics**
6. Students will have unprecedented access to the finest resources, experts and “Know-How” as well as the latest tools/equipment being used in Industry

©

The Very Best of Artificial Intelligence

The following Artificial Intelligence Course Material is novel in its unique approach to both students and faculty alike. The material is non-traditional in its presentation as it does not represent one specific book or follow a singular classical path. The Course Material is an aggregation of joint participations among top institutions, companies and organizations at the forefront of the “Best Practices in Artificial Intelligence.” More sound expertise is better.

Artificial Intelligence picked up significant speed and breadth in 2016-17, 2018 and 2019 due to innovation in GPU technology, Cloud Computing and Open-Source Software. As a result of computational horsepower and the demands on all cloud-based systems impacted by the Corona Virus Artificial Intelligence has accelerated by an order of magnitude more. This means the single classical methods of old are no longer enough to mobilize a properly prepared workforce.

This course is a carefully planned innovation in combining multiple sources, multiple medias, multiple experts and multiple forms of delivery to provide one of the finest, curated archives of resources, references and reading materials available to HCC Students.

We have sought the finest known resources & top name experts to present the “BEST of A.I.”

©

See How Far Things Have Come – “Assisted”

This Same Platform is Customized for Manufacturing
Nvidia DriveWorks Platform

Xavier AGX

NVIDIA DRIVE AGX Developer Kit - GTC Presentation

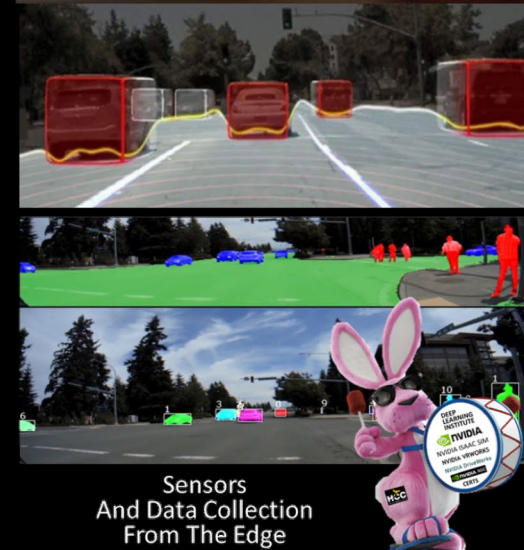


AUTOMATED DRIVING + AI COCKPIT

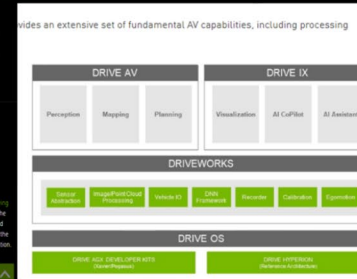
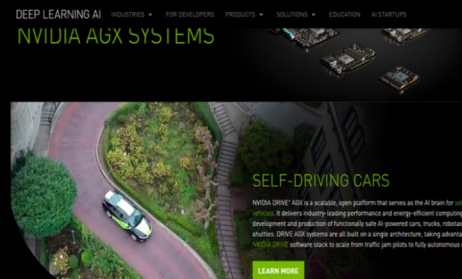


NVIDIA DRIVE CONSTELLATION

Virtual Reality Autonomous Vehicle Simulator



Sensors
And Data Collection
From The Edge



You will learn about the Platforms
Online - **LIVE**

5 of 70

The Nvidia Drive Platform using
Xavier AGX is the best “Edge” which is
also being used in Manufacturing

A.I. 1370



Intro A.I.
#2 of 16
Online

Videos, Blogs, Seminars and Expert Panels Methods and Techniques, Conferences, Code Notebooks, Streams and Contests, Dashboards, Audio and more

To Learn How, is To See How Experts do A.I.

A.I. tools and discussion now take many forms which we will introduce

©

A.I. is inherently multiple medias
Online - **LIVE**

6 of 70

A.I. is LIVE, Realtime and The Best.
Learn to Build an Education Strategy

A.I. 1371



Intro A.I.
#1 of 16
Online



CEO Jensen Huang



NVIDIA Keynote at Siggraph 2023

[NVIDIA Keynote at SIGGRAPH 2023 - YouTube](#)

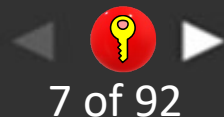
Please take the time to see this Keynote speech
This is a very important speech to all industries

“The Course MOTTO” About Material:

1. You don't need to be an expert – don't stress
2. You do need to know about Key Topics
3. You do need to be “Professionally Aware”
4. Consider yourself INFORMED at an Executive Level
5. Focus your attention on the “Gems” Slide 78ish for grading

©

You are at the beginning of a new
era in teaching machines to learn



A.I. is about THINKING and LEARNING
Let's dig in, this is an important talk.

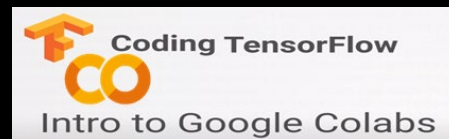
A.I. 1370



Intro A.I.
#2 of 16
Online



Use As Much As You Can



“Oh, there is too much material” is
the reason A.I. exists.

Job One in A.I. is to build tools and

Processes that help us

quickly make sense out of

TOO MUCH DATA.

In this course you will learn

how to leverage tools like:

Working in teams

Building a workflow for learning

And learning you do not need to

know it all – you need to know where

To locate it all...

©

Our job is to teach you – prepare you.

Online - LIVE

8 of 70



Your job is to make the most of it.

You can use as much or as little as you want

This is the basic Pilot 2D Map of A.I. Components (This simple version is necessary, now)



This is just a peek into how big A.I.
really is

9 of 70

- Basic A.I. Roadmap show the many Key areas of A.I.

The Very Best of You is learning to work Smart

The purpose of this course is to:

1. Show you how to learn deeper and faster
2. Present the finest available material by the Top Institutions
3. Show you how to continue to learn (learning never stops)
4. Introduce you to the Professional Tools and Environments
5. Get you into “Hand’s-on” forums, contests, challenges
6. Teach you how to ask great questions/trouble-shoot
7. Show you how to grow and direct your abilities

Do not be overwhelmed by the amount of material - you should use most of
It as a reference to learn and research more of what interests you

**Our job is to prepare you about how to use all the resources
and teach you where to look for the best information**

©



These are A.I. Embedded Templates

Templates with embedded A.I. can be personalized, customized and re-purposed for the audience, the instructor or the delivery system.

Course Templates with embedded A.I. are a “New Education Technology.”

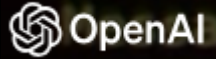
A Course is a Unit of Information

Scalability is essential

Everyone is potentially a Student

©

A.I. 1370



Note 02
#1 of 16

Introducing ChatGPT

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests.

[Try ChatGPT ↗](#)

[Read about ChatGPT Plus](#)

[OpenAI](#)

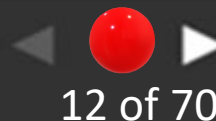
Courtesy OpenAI

Safety & Responsibility with A.I. starts now with all of us.

[Six Essential Elements Of A Responsible AI Model \(forbes.com\)](#)

©

Safety and Responsibility includes
all of us.



12 of 70

We want you to experiment –
Try out new things, see what they
can do

A.I. 1370



Week 01
#1 of 16

Welcome To HCC's Bachelor's Program In Artificial Intelligence & Robotics

The best way to learn about A.I. and all the topics associated is to begin at the very beginning. This is course one and it is terrific. It will help You understand the big picture of what is going on with A.I. We call this the "Executive View" – informing yourself well enough to make sense out of what you hear in the marketplace.

©

Module 01 Intro to A.I. – Course Setup



History, Theory and Platforms 01

Online - LIVE

13 of 70

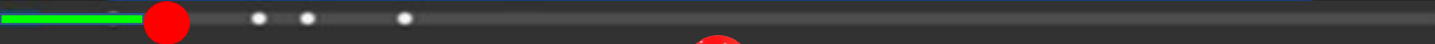
Welcome, You just selected the Right Course.

There isn't a more necessary topic on the planet today. A.I. will affect everyone and everything. Our goal in providing the course is to step up the pace in educating as many people as possible, as fast as possible about the key topics involved in Artificial Intelligence. In this beginning course we will introduce you to a "sampling" of the important subjects and how far things have come. We will be presenting the finest information from the very best Experts and Institutions like MIT, IBM, Nvidia, Stanford, CalTech, CMU, Microsoft, Google, Amazon and others. You will learn about the very "BEST PRACTICES."

We will go like the wind and provide you with as much information and training as you think you can hold. You should be taking this course if you: 1) LOVE the subject, 2) Need to find that high paying dream job, or 3) you need to KNOW the real truth about the topics that will truly change your business, the marketplace and how to get ahead of the curve in any industry before it is too late. Our job is to properly prepare you with the fundamentals so you can be quickly effective or continue to learn on a valuable path.

A.I. isn't new, but it has been highly accelerated to warp speed because of GPU's, COVID-19 and Cloud Computing. Right now, there are not enough people to do the jobs required to move A.I. into Healthcare, Cybersecurity, Oil and Gas, Telecommunications, Defense and Education quickly enough to meet the demand. We are counting on you! Buckle-Up!

©



This is the hottest topic on the planet   Congratulations. Very wise choice.

Online - LIVE

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Raising the Bar on Human Intelligence

One of our key goals with this course and the overall program is to help raise the level of HUMAN INTELLIGENCE prior to the use of A.I. As an A.I. professional we encourage you to THINK about the material you are learning. We don't want you to memorize the material. We want you to use the material, think about the subjects and reason out the issues in order to solve problems. A.I. is about solving problems and learning to think about your options, methods and results.

Lots of Work. Lots of Reward. This may be the first course you take because you really want to know. Consider this course a gift because it offers you insight into one of the most important subjects of our time. You are now a part of this A.I. process. We want you to be successful from the human intelligence perspective, first. Think, reason, plan, review, understand as much meaning as possible.

The quality of questions you learn to ask is far more important than seeking single answers. Please Go for It! **The following is the daily breakdown of the course.**

©

Artificial Intelligence Course 1



Introduction to A.I.

Professor Brown



World-Class Authorities and Topics ☐ Fascinating and Complete
Online - LIVE 1 of 40 "State-of-the-Art" 96.00

Artificial Intelligence Course 1



Games, Prelude to A.I.

Professor Brown



Fundamentals of Best Practices in A.I. ☐ Games, Prelude to A.I. 03
Online - LIVE 1 of 40 96.00

Artificial Intelligence Course 1



Machine Learning/Deep Learning

Professor Brown

A.I. is all about Learning from the Data ☐ Building the Data Pipeline
Online - LIVE 1 of 40 Neural Networks 96.00

Artificial Intelligence Course 1



Computer Vision

Professor Brown



Teaching Cameras and Sensors to See ☐ What is a Camera, a Lens and Vision
Online - LIVE 1 of 40 96.00

Artificial Intelligence Course 1



Natural Language Processing

Professor Brown



Learn about BERT, JARVIS and Watson ☐ Understanding, Ambiguity, Meaning
Online - LIVE 1 of 40 96.00

Artificial Intelligence Course 1



Robotics

Professor Brown



Walking, Rolling, Crawling, Flying, Swimming ☐ Hardware, software, Simulation
Online - LIVE 1 of 40 96.00

Artificial Intelligence Course 1



Platforms and Security

Professor Brown



A.I. in Cybersecurity is the #1 Issue ◀ ▶ Final Word on Platform Types
Online - **LIVE** 1 of 40 96.00

Artificial Intelligence Course 1



XXtras

Professor Brown



The Xxtras are open exploration ◀ ▶ We will squeeze in Xxtras like
Online - **LIVE** 1 of 40 Communications and Skills++

A.I. 1370



#1 of 16
Online

Course 1 History, Theory and Platforms

You are here ★ **Course 1 - Module 1 - Introduction to A.I. - Course Setup**

Course 1 - Module 2 - Introduction to A.I. – The Big Names, Big Issues

Course 1 - Module 3 - Games, Prelude to A.I.

Course 1 - Module 4 - Games Change Everything

Course 1 - Module 5 - Machine Learning – The Data

Course 1 - Module 6 - Machine Learning – The Pipeline

Course 1 - Module 7 - Deep Learning – Neural Networks

Course 1 - Module 8 – Deep Learning – Real Learning Good Analytics

Course 1 - Module 9 - Computer Vision – Image Processing

Course 1 - Module 10 - Computer Vision – Image Understanding

Course 1 - Module 11 - Natural Language Processing – Basics

Course 1 - Module 12 - NLP – Speech Does it All

Course 1 - Module 13 - Robotics – Grand Tour

Course 1 - Module 14 - Robotics – Hard and Soft Issues

Course 1 - Module 15 - Platforms and Security – Setups

Course 1 - Module 16 - Platforms and Security – Cognitive Issues

Course 1 - Module 17 – Xxtras – Communications and Skills

©

History, Theory and Platforms

Online - **LIVE**



17 Modules for Course 01 A.I.

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Reality is taking a look at the A.i. Jobs

On Tuesday, career and job site LinkedIn released its annual “Emerging Jobs” list, which identifies the roles that have seen the largest rate of hiring growth from 2015 through this year. No. 1 on the list: Artificial Intelligence Specialist — typically an engineer, researcher or other specialty that focuses on machine learning and artificial intelligence, figuring out things like where it makes sense to implement AI or building AI systems.

Hiring for this role has been tremendous, growing 74% annually in the past 4 years alone. **“AI has infiltrated every industry, and right now the demand for people skilled in AI is outpacing the supply for it,”** Guy Berger, the principal economist at LinkedIn, tells MarketWatch. “This is the third year in a row a role related to machine learning or artificial intelligence has topped the list, and we can only expect demand to increase.”

The pay is impressive too, with AI roles often commanding six figures. Jobs site Indeed notes that artificial intelligence engineers in San Francisco, for example, rake in \$120,000 to upwards of \$160,000. Sometimes AI roles can garner pay of \$250,000 or more. LinkedIn isn't the only company to highlight an AI specialty role as a job to watch. Indeed's annual list of the “25 best jobs of 2019” named machine learning engineer as No. 1, citing a 344% increase in job postings in the past few years and an annual base salaries of \$146,000, among other perks.

So what's behind this rapid growth in AI jobs? Berger says that “almost everyone” is hiring for these roles from the obvious (tech and automotive) to the more surprising (higher education and sports).

And these offer a real opportunity even for people who aren't currently in AI: “We're in an extremely tight labor market so companies are really looking to hire whoever can get the job done,” says Berger — who notes that learning skills like TensorFlow and Python, as well as diving into machine learning and natural language processing, could help you land the role. You can often take these kinds of classes as certificate programs from local universities, coding schools and more.

Rounding out the top 5 jobs on LinkedIn's emerging jobs report are robotics engineers (40% annual hiring growth), data scientist (37%), full stack engineer (35%) and site reliability engineer (34%). **“While many of these jobs are tech roles, they're not necessarily in the tech industry.** Every company has had to embrace tech at some level and we're seeing that reflected in these high-growth jobs,” adds Berger.

But interestingly, there are also a number of client-facing roles that are experiencing rapid hiring growth, such as customer success specialist and sales development representative. Many roles like this “are heavily reliant on relationships, so being skilled in things like communication, problem-solving and collaboration are key,” Berger notes, adding that for these kinds of gigs companies “will rely on people skills that can't be automated, successfully complementing new technologies.”

(29) The Role of AI in 2023 | LinkedIn

©

A.I. is in every field – universally applied

Online - **LIVE**

19 of 70

If you can do fantasy football you
can do A.I.

A.I. Jobs

**U.S. companies are on a hiring spree for A.I. jobs—
and they pay an average of \$146,000 – CNBC Story
Jul 2023**

[U.S. companies are on a hiring spree for A.I. jobs that pay an average
of \\$146,000 \(cnbc.com\)](#)

Tech's biggest companies are placing huge bets on artificial intelligence (Warning: may be paywalled; alternative source)," reports the New York Times, and "typical AI specialists, including both Ph.D.'s fresh out of school and people with less education and just a few years of experience, can be paid from **\$300,000 to \$500,000** a year or more in salary and company stock."

[Tech Giants Are Paying Huge Salaries For Scarce AI Talent - Slashdot](#)

This is a tiny beginning of what is coming

The field of A.I. is rapidly opening
many different types of jobs

A.I. 1370



#1 of 16
Online

Here's What We Are Going to Do...

A Little History

Some Videos

A Little Theory

A Few Definitions

TimeLine

Platforms

Glossary Project

Syllabus and Scoring

Setup Your Team

Laboratory Overview

Term Projects/Challenges (IBM and Kaggle)

Weekly Leadership Skills

[\(29\) Top 5 In-Demand Tech Skills For Jobs In 2023 | LinkedIn](#)

©

This is a quick Syllabus of the basics



Agenda for the Course Modules

Online - LIVE

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04:00

Building a Team in A.I. is Crucial

Four core types of AI talent – it is about Problem Solving from different perspectives

Collins sees four core disciplines as necessary for AI teams:

- Someone who understands the business processes critical to establishing real-world scenarios and valuable outcomes.
- Someone who understands analytics – machine learning, statistics, forecasting and optimization – that leads to using the right techniques.
- Someone who understands data: Where does it come from? What is the quality? How should it be handled to preserve security and trust?
- The final "someone" is an AI architect who knows how to put analytics in action by operationalizing for outcomes.

"Those disciplines encourage understanding the scientific process and thinking in terms of complex, interactive systems," Collins says. "They're often good at the critical thinking skills required to establish good experiments and the outcomes of applied machine learning."

Keith Collins CIO SAS

©

Team Diversity Leads to Innovation

The value of a diverse AI team

The value of a diverse team ranges far and wide: It can help your organization better combat against AI bias, for instance. It's also important to solving business problems – including your largest and toughest problems – which is presumably one reason you're developing an AI strategy in the first place.

"It is generally accepted that diversity of opinions is critical to all complex problem-solving," says Jeff McGehee, senior data scientist and IoT practice lead at Very. "Diversity is all about life experience, and professional background is a large part of most individuals' life experience, which can add dimension to AI projects and provide new perspectives to finding innovative solutions."

McGehee also points out that building diverse teams – AI or otherwise – requires an active effort on the part of your company as part of recruiting and hiring practices. Sitting back and assuming diversity will find you is not a viable team-building strategy.

[**Related read:** Why diverse IT teams have a competitive edge.]

With that in mind, let's look at a range of experts and roles – including non-technical roles – that can be valuable to an AI team.

JEFF McGehee Senior Data Scientist and IOT Lead at VERY

©

A.I. 1370



Diagram
#1 of 16
Online

Building Your Team

Team Name: _____ (A.I. Related)

Team-Mates: _____ email: _____
_____ email: _____
_____ email: _____

Team Motto: _____

Team Song: _____

Team Strengths: _____

Team Picture: _____

©

Use this form with The Team Meeting



Building Your "Super-Team"

Online - LIVE

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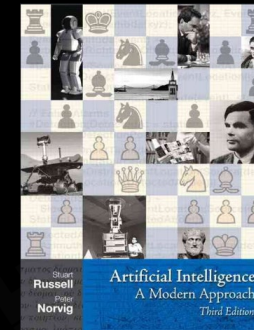
Term

A Little History

History – What is Artificial Intelligence?

In “**Artificial Intelligence: A Modern Approach**”, **Stuart Russell** and Peter Norvig **defined AI** as “the designing and building of **intelligent** agents that receive percepts from the environment and take actions that affect that environment.”

In general use, the term "artificial intelligence" means a program which mimics human cognition. At least some of the things we associate with other minds, such as learning and problem solving can be done by computers, though not in the same way as we do.^[1] Andreas Kaplan and Michael Haenlein define AI as a system's ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation and actions.^[2]



1. Patrick Winston (**Open Courseware at MIT**) <https://www.youtube.com/watch?v=TjZBTDzGeGg>

These are being touted as the 3 basic types of Artificial Intelligence

1. **Artificial Narrow Intelligence (ANI)**, narrow range of abilities – task specific – automated processes
2. **Artificial General Intelligence (AGI)**, which is on par with human capabilities; independent of humans
3. **Artificial Superintelligence (ASI)**, which is more capable than a human.

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A.I. 1370



#1 of 16
Online

Sentience is the capacity to feel, perceive, or experience subjectively. Eighteenth-century philosophers used the concept to distinguish the ability to think (reason) from the ability to feel (**sentience**). Wikipedia



Courtesy AMC Studios

AMC Series about Artificial intelligence and Robotics, focusing on the social, cultural, psychological impact of the invention of anthropomorphic robots called "synths".

The series highlights the subjects of consciousness and what it means to be human.

©

<https://www.youtube.com/watch?v=BV8qFeZxZPE>



Online - LIVE AMC Studios, Channel 4 26 of 70

Lars LUNDSTROM, Real Humans
provided the basis for ethics of A.I.
The Ethics of A.I.

Let's Talk About Consciousness, briefly

Philosophy and Sentience^[edit]

In the philosophy of consciousness, *sentience* can refer to the ability of any entity to have subjective perceptual experiences, or as some philosophers refer to them, "qualia"^[3] -- in other words, the ability to have states that it *feels like something* to be in.^[4] This is distinct from other features of the mind and consciousness, such as creativity, intelligence, sapience, self-awareness, and intentionality (the ability to have thoughts about something). Sentience is a minimalistic way of defining *consciousness*, which otherwise commonly and collectively describes sentience plus further features of the mind. These further features of consciousness may not be necessary for sentience, which rests on the capacity to feel sensations and emotions.^[5]

Some philosophers, notably Colin McGinn, believe that the physical process causing sentience to happen will never be understood, a position known as "new mysterianism." They do not deny that most other aspects of consciousness are subject to scientific investigation, but they argue that subjective experiences will never be explained; i.e., sentience is the *only* aspect of consciousness that cannot be explained. Other philosophers (such as Daniel Dennett) disagree, arguing that all aspects of consciousness will eventually be explained by science.^[6] Wikipedia

©

Let's Talk About Consciousness, briefly

What is The Thousand Brains Theory of Intelligence?

Underlying our machine intelligence work is scientific research that demonstrates how the brain creates intelligence. In October 2018 we released a major new theory for intelligence and cortical computation. We have several resources available for you to learn more.

Peer-reviewed research paper introducing the theory: A Framework for Intelligence and Cortical Function Based on Grid Cells in the Neocortex

Companion paper that describes the theory in non-scientific terms: Companion to A Framework for Intelligence and Cortical Function

Jeff Hawkins on Lex Fridman's Artificial Intelligence (AI) Podcast: Thousand Brains Theory of Intelligence

Video Presentation: The Thousand Brains Theory: A Framework for Understanding the Neocortex and Building Intelligent Machines

New York Times story on co-founder Jeff Hawkins and our work: Jeff Hawkins Is Finally Ready to Explain His Brain Research

https://www.youtube.com/watch?time_continue=3&v=-EVqrDIAqYo&feature=emb_logo

©

Neuro-Cognitive A.I. Numenta

We are one of the few teams that has developed large scale theories of the brain that are biologically constrained, testable, and implemented in software. We believe The Thousand Brains Theory will be foundational to the creation of truly intelligent systems.

How Our Research Will Impact the Future of Machine Intelligence

Over the past decade, machine learning and AI have succeeded at performing many tasks that were unimaginable earlier. Problems such as identifying a cat in an image or recognizing someone's speech are now routine in our lives. Yet, the way these technologies accomplish the tasks is fundamentally different than how our brain accomplishes the same tasks. For example, a machine learning system that recognizes a cat may be trained with over a million labeled training images in order to reliably identify a cat, whereas a human child learns about a cat with just a handful of examples. Moreover, that child knows that the cat may purr, the cat may scratch, and the cat can jump high, whereas all the machine knows is what a cat looks like. The machine doesn't understand cat; the child does.

The limitations faced by today's AI systems are generally agreed in the machine learning community. Such systems require enormous time and resource to train, are brittle to noise, cannot learn continuously, and do not generalize but rather accomplish narrow, specific goals. We are still a long way from a robot performing tasks that a child can handle easily.

Courtesy Numenta

©



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#1 of 16
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The chief executive of the young Canadian company Interaxon, Garten sought to use brain-controlled interface technology — a science fiction-like development that allows a device to be controlled by the brain's electrical activity — in a wearable device. With Google's Glass and Nike's FuelBand leading the way, so-called wearables are expected to grow from an \$8 billion (in annual revenue) market today to a \$20 billion one by 2017, according to Future-source Consulting. Using brainwaves to control them seemed like, well, a no-brainer.

©

Neuro-Cognitive A.I. devices

Online - LIVE



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EEG Brain Signals, Machine Learning to Control Sphero 2.0

Keywords: EEG, Machine Learning, EmotivBCI Insight NodeJS, Sphero2



(Source: [Emotiv Insight](#))

(Source: "[Sphero](#)" by ChrisGoldNY)



Emotiv Insight



MindFlex
Game

©

A.I. 1370
Zewde



#1 of 16
Online

Human Intelligence

The Alan Turing Institute

[Home](#) | [The Alan Turing Institute](#)

Home + Events

Turing Lecture: Is education AI-ready?

The first in The Turing Lecture mini-series exploring the role of AI and data science in our lives post-lockdown

[Learn more](#) ↓ [Register now](#)

Add to Calendar 07/29/2020 03:00 PM 07/29/2020 04:30 PM Europe/London Turing Lecture: Is education AI-ready? Location of the event

Wednesday 29 Jul 2020
Time: 15:00 - 16:30

Event agenda ↓

Event type
[Lecture](#)

Audience type
[General](#)

Free

Event series
[Turing Lectures](#)

The Alan Turing Institute is a great Resource of ongoing Lectures

©

The Alan Turing Institute Webinars

Online - **LIVE**

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Professor Rose Luckin, Turing Institute
Is Education A.I. Ready?

A.I. 1370



OWC 01
#1 of 16
Online



Massachusetts
Institute of
Technology



6.034, Fall 2010

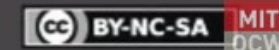
Artificial Intelligence

Patrick H. Winston

Lecture 1: Introduction and Scope

Patrick Winston (Open Courseware at MIT) <https://www.youtube.com/watch?v=TjZBTDzGeGg>

MITOPENCOURSEWARE
MASSACHUSETTS INSTITUTE OF TECHNOLOGY



You will hear from the Very Best Experts in the World

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Begin with the Definition of A.I.



Speaker 01 Patrick Winston OCW

Online - **LIVE**

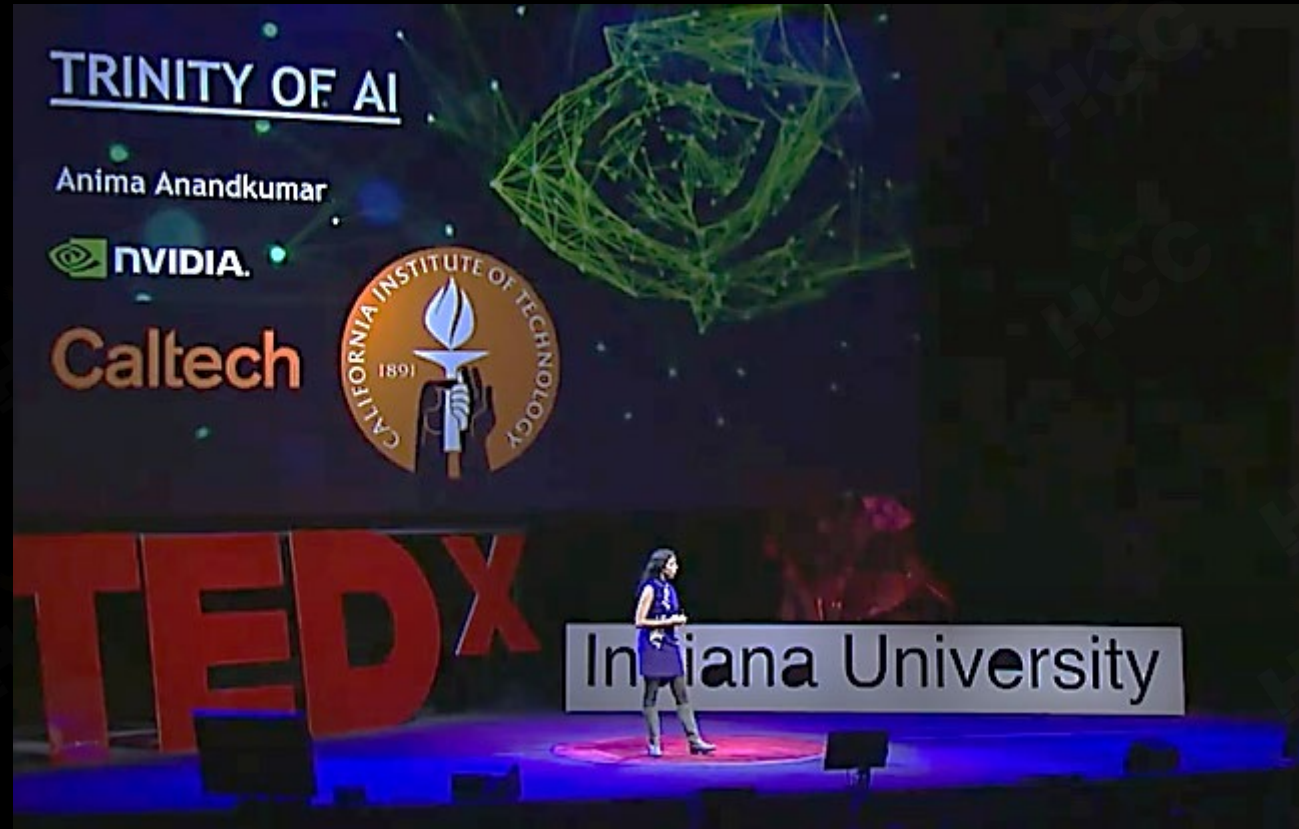
33 of 70

47:18

A.I. 1370



#1 of 16
Online



<https://www.youtube.com/watch?v=zkJ5IEuGQDw>

©

“The Trinity of A.i.” is best overview
Online - LIVE

◀ ● ▶
34 of 70

Dr. Anima Anandkumar
Tensors

00.18

A.I. 1370

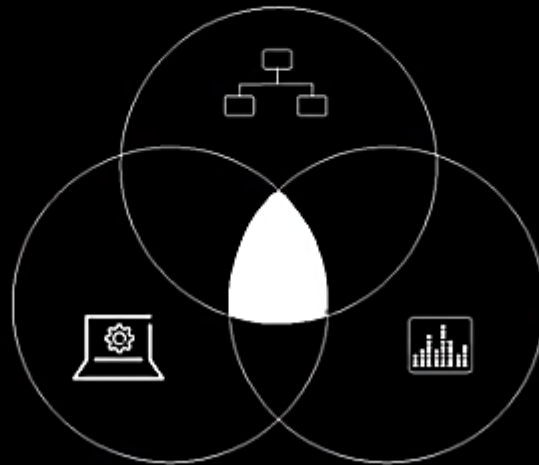


#1 of 16
Online

TRINITY OF AI

ALGORITHMS

COMPUTE



DATA

<https://www.youtube.com/watch?v=zkJ5IEuGQDw>

©

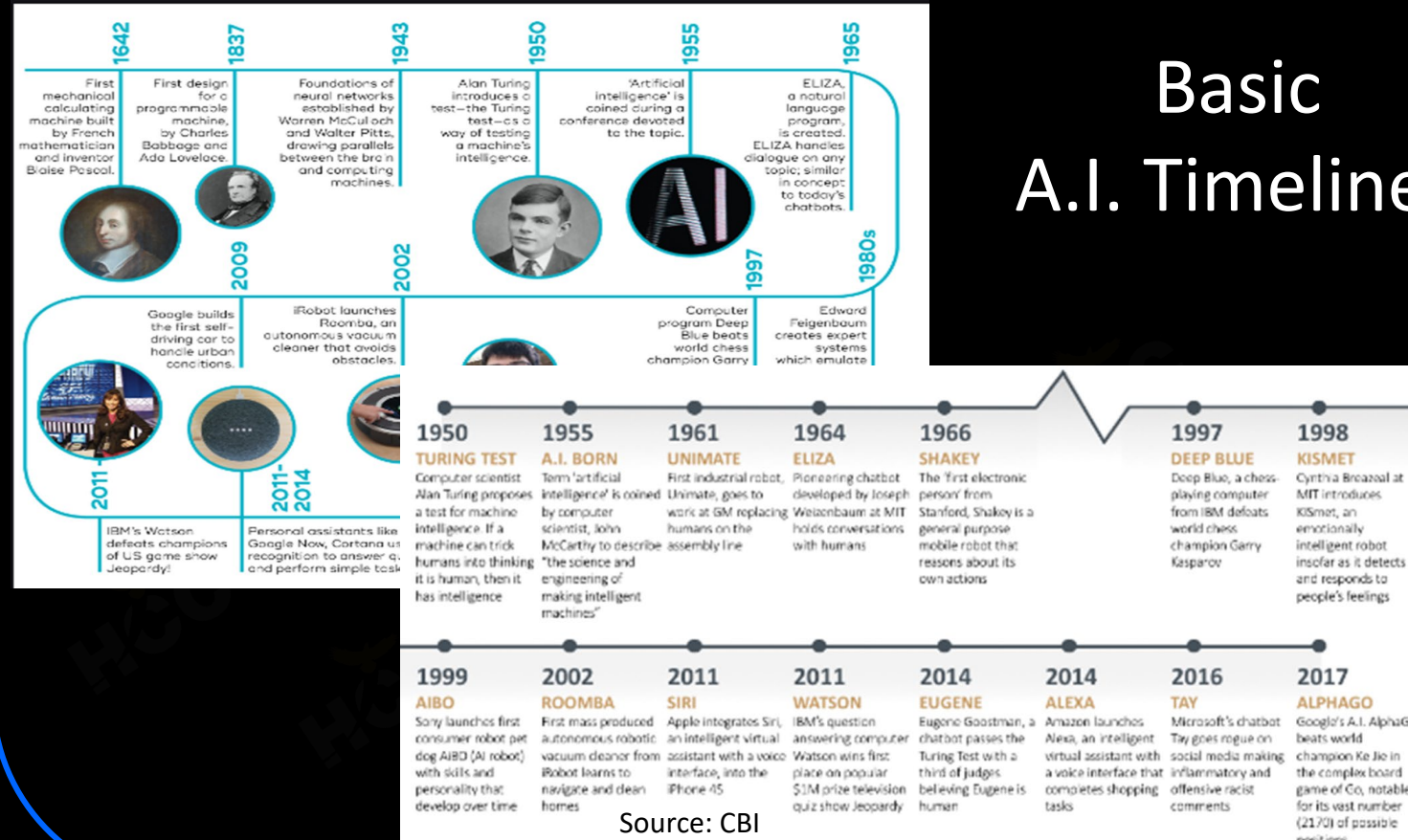
The Data is King "The Trinity of A.i."
Online - LIVE

35 of 70

Dr. Anima Anandkumar's Work

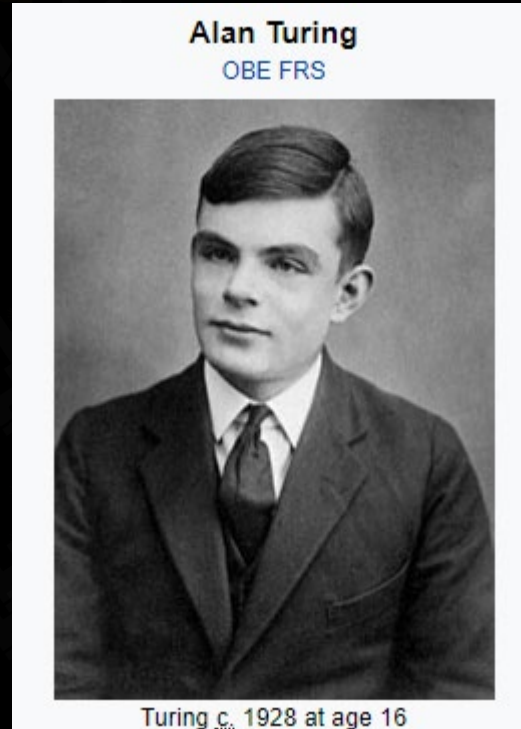
00.18

Basic A.I. Timelines



Source: CBI

©



The Father of A.I.

Alan Mathison Turing OBE FRS (/ˈtʃʊərɪŋ/; 23 June 1912 – 7 June 1954) was an English^[6] mathematician, computer scientist, logician, cryptanalyst, philosopher, and theoretical biologist.^[7] Turing was highly influential in the development of theoretical computer science, providing a formalization of the concepts of algorithm and computation with the Turing machine, which can be considered a model of a general-purpose computer.^{[8][9][10]} Turing is widely considered to be the father of theoretical computer science and artificial intelligence.^[11] Despite these accomplishments, he was never fully recognized in his home country during his lifetime due to his homosexuality and because much of his work was covered by the Official Secrets Act.

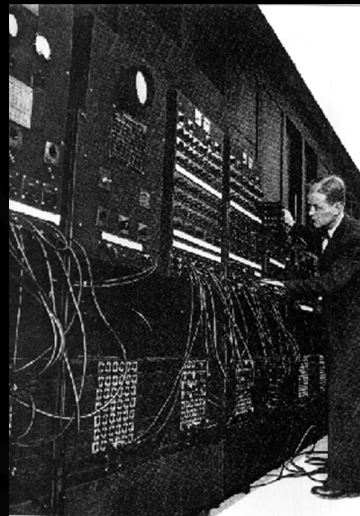
During the Second World War, Turing worked for the Government Code and Cypher School (GC&CS) at Bletchley Park, Britain's codebreaking center that produced Ultra intelligence. For a time he led Hut 8, the section that was responsible for German naval cryptanalysis. Here, he devised a number of techniques for speeding the breaking of German ciphers, including improvements to the pre-war Polish bombe method, an electromechanical machine that could find settings for the Enigma machine.

Wikipedia Example

©

Where were the first Computers Being Used?

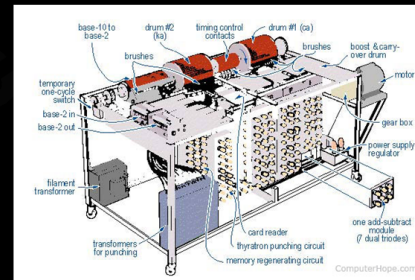
Post War and Post Princeton Von Neuman



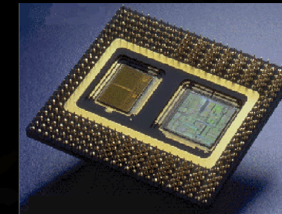
Eniac Computer UPENN
1946-55
Vacuum Tube

Courtesy Princeton

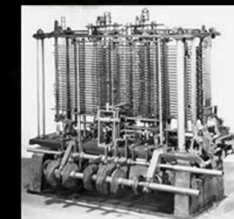
Census
Weather
Military



Atanasoff or ABC Computer
1937-42 Iowa State



Transistor (1947) and
Integrated Circuit (1958)
changed everything



Charles Babbage
1837

Today Quantum Computers are as real as an iPhone

©

A.I. 1370



ML 01
#1 of 16
Online



Deep Learning: <https://www.youtube.com/watch?v=Bo8MY4JpiXE> IPFCONLINE [PIERRE PINNA](#)

François Chollet: Keras, Deep Learning, and the Progress of AI | Lex Fridman Podcast #38

©

Machine Learning is about training



Machine Learning Pipelines

Online - LIVE

39 of 70

4 Hours

Machine Learning and Image Understanding



Courtesy DataScienceCentera;

Chihuahua or muffin?

<https://www.datasciencecentral.com/profiles/blogs/what-tomorrow-s-business-leaders-need-to-know-about-machine>

©

Robotics Need On-board Intelligence

Rodney Brooks MIT: <https://www.youtube.com/watch?v=C9p8B7-5MTI>

Kismet: <https://www.youtube.com/watch?v=8KRZX5KL4fA>

Kismet: <https://www.youtube.com/watch?v=ZJCiD5Smm4k>

Bristol Labs: <https://www.youtube.com/watch?v=Vwg7e-W7KZw>

Boston Dynamics: Atlas <https://www.bostondynamics.com/atlas>

Spot: <https://www.bostondynamics.com/spot>

Biology Labs: <https://www.youtube.com/watch?v=Ki-haVUrzNI>

Genetic Testing: <https://www.youtube.com/watch?v=SH4q428jAEs>

Disease Testing: <https://www.youtube.com/watch?v=ra0e97Wiqds>

Hod Lipson: https://www.ted.com/talks/hod_lipson_building_self_aware_robots?language=en

©

Robots will not be made without A.I. ◀ ▶ Robots are the embodiment for A.I.

Online - LIVE

41 of 70

4 Hours

A.I. 1370



#1 of 16
Online



So-called 'AI anchors' first appeared in China in November 2018, and there are now reported to be three AI anchors on-air. There's another in Thailand, reports of two more in Russia, and another slated to debut in the UAE.

<https://www.youtube.com/watch?v=5iZuffHPDAw>

©

Now there are fake models, presenters, etc



This is the real culprit behind Fake News

Online - LIVE

42 of 70

4 Hours

A.I. 1370



#1 of 16
Online

Deep fake sports presenter who can deliver video match reports



<https://www.dailymail.co.uk/news/article-8006223/Reuters-unveils-deep-fake-sports-presenter-deliver-video-match-reports.html>

©

Generative Facial Maps keep it real



Soccer Broadcast with News Avatar

Online - LIVE

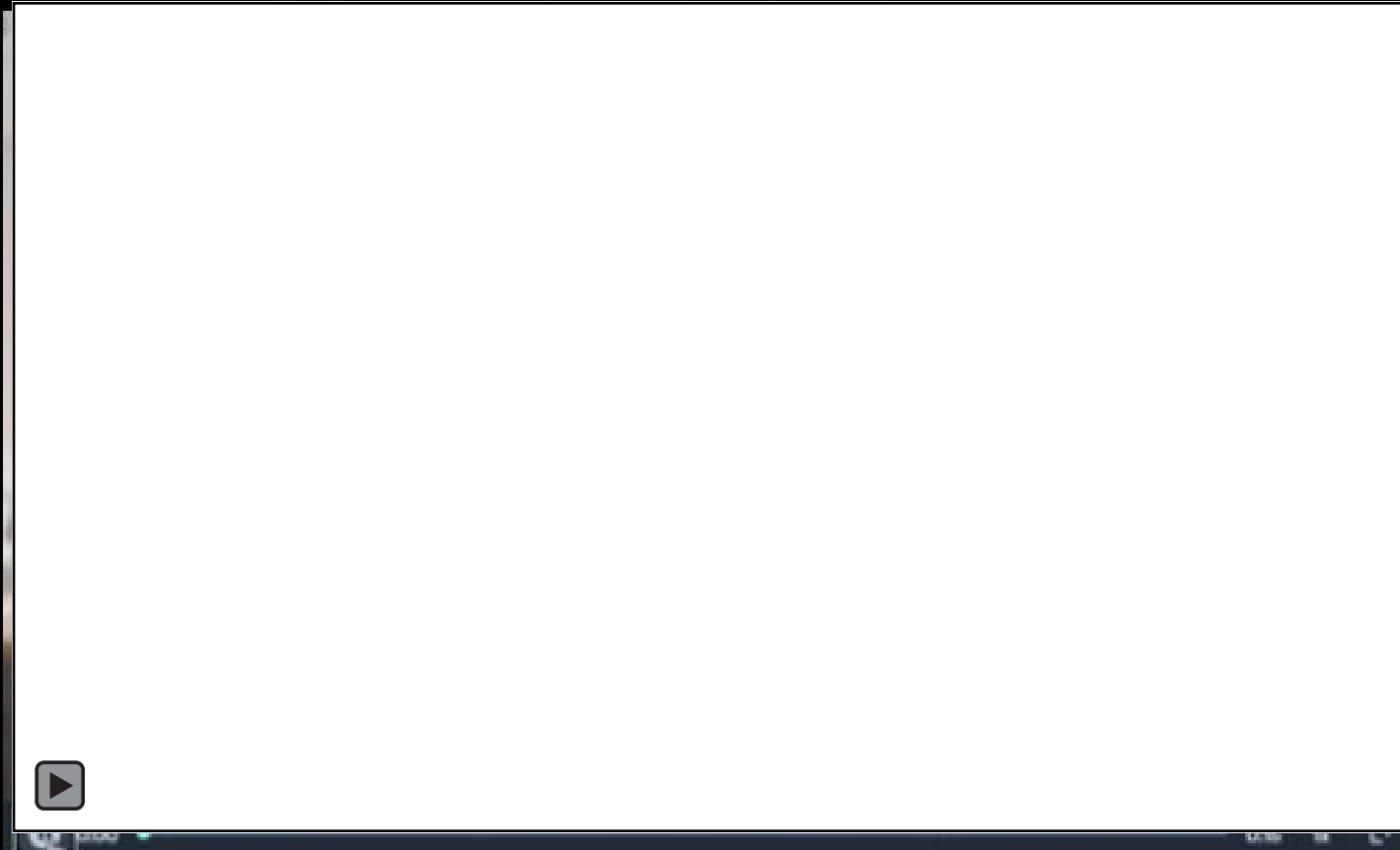
43 of 70

4 Hours

A.I. 1370



#1 of 16
Online



Real Synthetic Anchors/Avatars

©

This is "Am-I-Real," a synthetic avatar



Building Smart Avatars takes time

Online - **LIVE**

44 of 70

4 Hours

Student Life Communities could turn meetings into “Focused Expertise Team Discussion Forums”
Better known as “Birds of a Feather” Sessions



AM – I REAL

SYNTHETIC NEWS CO-ANCHOR



CAIO DE ARAGON CRUZ

REAL STUDENT NEWS CO-ANCHOR



STUDENT LIFE NEWS TEMPLATES ALLOW FOR CO-ANCHORS, INTERVIEWS, ONLINE ZOOM CHATS

There are many pros and cons to having this ability. In essence How would you best use this capability?

A.I. 1370



#1 of 16
Online

Doug Roble | TED2019

Digital humans that look just like us

DigiDoug



https://www.ted.com/talks/doug_roble_digital_humans_that_look_just_like_us/transcript?language=en#t-91910 <http://caisplusplus.usc.edu/>

©

Generative Human Models/Faces

Online - LIVE



46 of 70

<https://www.youtube.com/watch?v=piJ4Zke7EUw>

Digital Emily Project - USC

A.I. 1370



#1 of 16
Online



Courtesy Nvidia

Raytracing with A.I. will change everything further

©

Raytracing is changing AEC/Construction



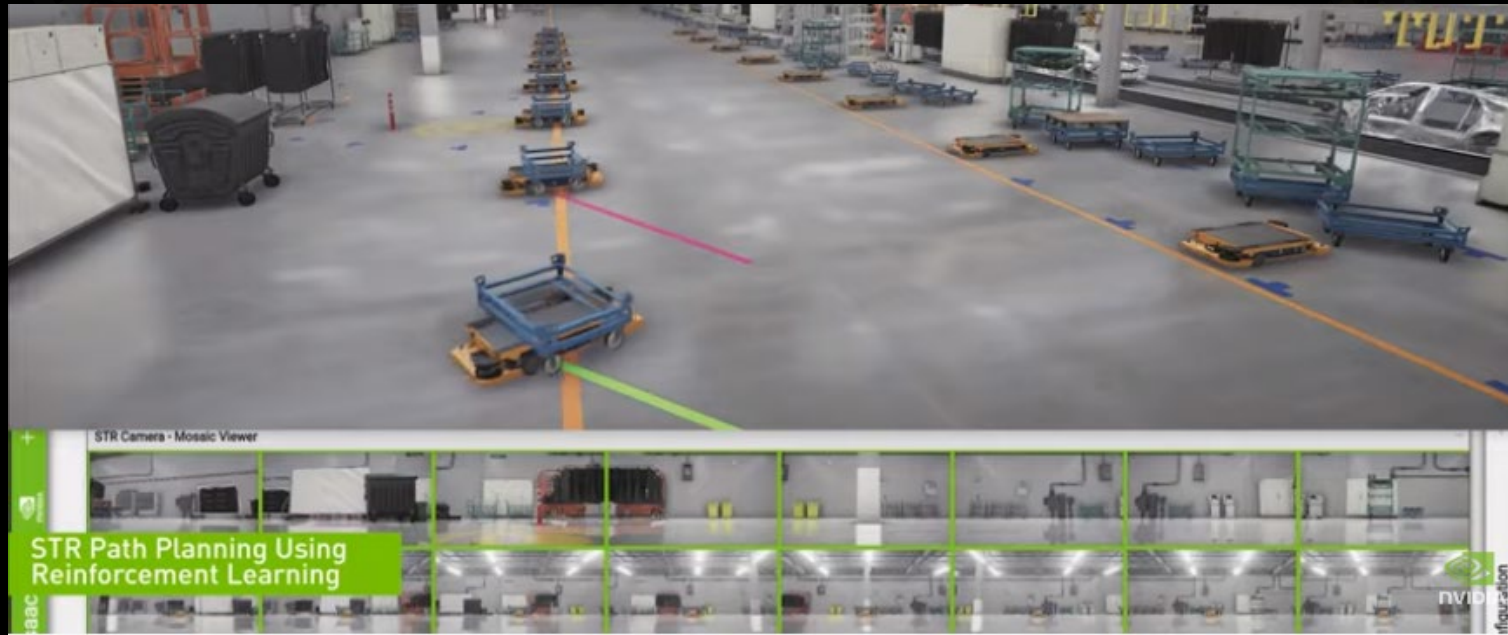
Rendering as a Service on a Blockchain

Online - LIVE

47 of 70

Future Netflix might be 1,000's of pieces

New Era of Simulation is quite Real



https://www.youtube.com/watch?v=v_uIHMfDUco

©

Simulation is so good you can plan and do

Online - **LIVE**

48 of 70

The lines between real-time simulation
are in lock step with Real-Time Real Ops

Realtime Path Navigation – Nvidia Drive Lab



Courtesy Nvidia Drive



<https://www.youtube.com/watch?v=ftsUg5VlzIE>

©

AR, Simulation and Control 360 degrees



The Nvidia Drive Platform is Special
Autonomous Machines

Online - **LIVE**

49 of 70

A.I. Includes These Subjects

Data Science, Data Curation and Big Data

Computer Science (hardware, software and Operating Systems)

Networking and Virtualization

Cyber-Security and Blockchain

Cloud Computing

Computer Vision and Image Recognition

Natural Language Processing (NLP)

Programming

Databases

Knowledge Graphs

Neural Networks and Decision Systems

Mathematics, Probability, Statistics, Logic

Robotics and Control Systems

IOT and IIOT

Simulation

Algorithm Development

Autonomous Machines

Learning Agents/Bots

Machine Learning

Deep Learning

Inference and Perception

Cognitive Development

Philosophy, reasoning/action

Information Distribution

Information Theory

Raytracing

Communications

Edge Computing

©

Most of these topics work together ◀ ▶ All of these are important

Online - **LIVE**

50 of 70

A.I. also Includes These Subjects

Logic,
Probability,
Continuous mathematics,
Perception,
Reasoning,
Learning, and action;
and everything from
Microelectronic Devices
to Robotic Planetary Explorers

©

You can Do What You Can Process



It's All About The Data

Code and Processor are tuned to the Data



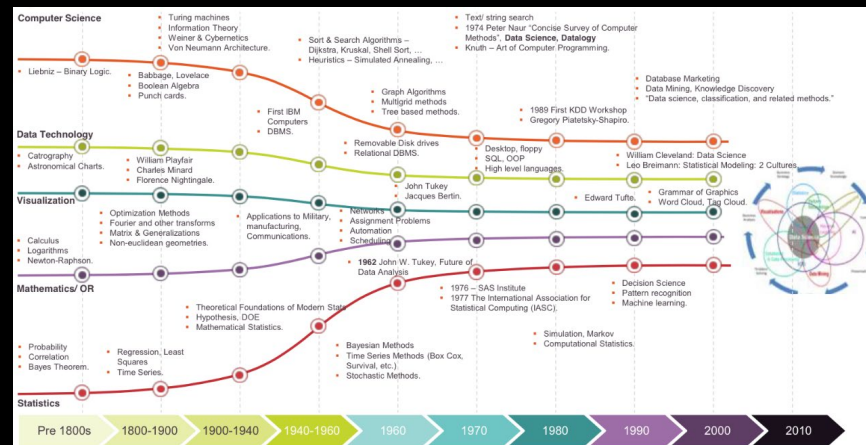
▶ This is the heyday of Data Science

Online - **LIVE**

24 of 70

4 Hours

Let's Talk About The Data Science



It is only now we are truly beginning to understand and value data

Human Intelligence

Whilst people are undoubtedly becoming more test-wise and are picking up specialized cognitive skills, as evidenced by studies such as the one conducted by Dr. Staff and colleagues," Woodley said in an email to The Huffington Post, "they are unfortunately not becoming more innovative, or better complex problem solvers." (the keys to A.I.)

Michael Woodley

©

Let's Talk About The Data

90% of the world's data has been created in just the past few years. Faced with overwhelming amounts of data, organizations are struggling to extract the powerful insights they need to make smarter business decisions. To help uncover the true value of your data, MIT Institute for Data, Systems, and Society (IDSS) created the online course *Data Science and Big Data Analytics: Making Data-Driven Decisions* for data scientist professionals looking to harness data in new and innovative ways.

Gartner defines big data as the three Vs: high-volume, high-velocity, high-variety information assets. While all three Vs are growing, variety is becoming the single biggest driver of big-data investments, as seen in the results of a recent survey by New Vantage Partners. This trend will continue to grow as firms seek to integrate more sources and focus on the "long tail" of big data. From schema-free JSON to nested types in other databases (relational and NoSQL), to non-flat data (Avro, Parquet, XML), data formats are, multiplying and connectors are becoming crucial. Companies will continue to evaluate analytics platforms based on their ability to provide live direct connectivity to these disparate sources.

<https://idss.mit.edu/about-us/>

©

A.I. 1370



#1 of 16
Online

Everything is About Data, even Joke of the Day.



Why did the Robot marry his partner?

Because he couldn't resistor.

©

<https://www.youtube.com/watch?v=DLIU5tC3LAs>

Online - LIVE

55 of 70

Even Jokes require rich semantics
Lieutenant Commander Data Feels Laughter

A.I. 1370



#1 of 16
Online

Let's Talk About Data and Joke of the Day



Courtesy the Punchline

The Punchline 7 years later

©

https://www.youtube.com/watch?v=d1N5w9q_7o8



Ambiguity is King in the Joke World

Online - LIVE

56 of 70

4 Hours

A.I. 1370



#1 of 16
Online

Joke of the Day.



Why did the chicken cross the road?

To discuss NLP specifics about context and storytelling

©

Lorenzo Ampil Dec 26, 2019

Online - LIVE



MIT Teaching an AI to Tell Jokes

57 of 70

Joke of the Day – Yes that is an A.I. Topic

This model, named "Tito Joker", was built with the goal of creating an AI that understands humor well enough to tell jokes that are actually funny. The code is fully open-sourced on [github](#).

It was trained by fine-tuning the recently released [OpenAI GPT-2](#) on an open sourced [jokes dataset](#).

Why is the model named Tito Joker? Because in Filipino, "tito" means "uncle" when translated to English, and in the Philippines, we all have that uncle who says the corniest jokes!

Acknowledgments

Special thanks to [Hugging Face](#) for their implementation of OpenAI GPT-2 using PyTorch and [Thinking Machines Data Science](#) for sponsoring the server that I am running Tito Joker on.

Ask [Tito Joker](#) anything! :)

Created by [Lorenzo Ampil](#).

Ask any question

Why did the chicken cross the Road?

<https://www.kaggle.com/abhinavmoudgil95/short-jokes>

©

What We Mean by A.I. Platforms

IBM is one of the best AI engines because of Watson. It comes powered by modern innovation in machine learning to allow the models to learn more with less data. Developers can choose to build new models from scratch or use Watson APIs and pre-trained solutions to power existing applications.

What makes IBM Watson unique is its ability to learn from small data sets. IBM believes that it's the quality of data, rather than quantity, which makes the difference.

IBM Watson comprises several enterprise-grade AI services, applications, and tooling. These include Watson Assistant, Watson Studio, AI OpenScale, Watson Discovery, Natural Language Understanding, Discovery News, Knowledge Studio, Language Translator, Natural Language Classifier, Personality Insights, Tone Analyzer, Visual Recognition, Speech to Text, and Text to Speech.

Additionally, Watson allows enterprises to integrate its services into Salesforce and Box. The Salesforce integration helps enterprises to provide AI-powered solutions and make quick and smart decisions across service and sales. Whereas, the Box integration is aimed to automate the structure of content, unlock hidden value, and automate workflows in the cloud. Source: IBM



©

Ai Platforms: IBM, Microsoft, Amazon, Google, Nvidia

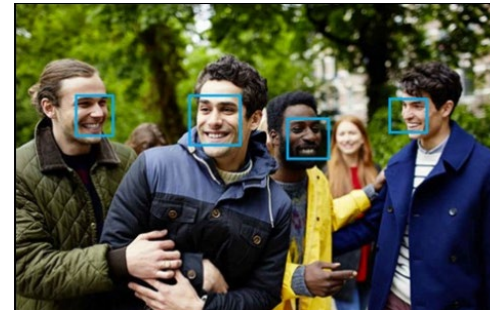
1. Microsoft Cognitive Services

Microsoft provides intelligent APIs which developers can use to infuse vision, speech, language and knowledge capabilities into applications, websites, and bots.

Part of the Microsoft AI platform, the **Microsoft Cognitive Services** allows developers to compose intelligent applications, that can be customized according to the availability, security and compliance requirements of the organization.

Microsoft Cognitive Services include the following:

- **Emotion API**
- Computer Vision API
- **Language Understanding (LUIS)**
- Speaker Recognition API



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PS01
#1 of 16
Online

Even Puzzle of The Day is About The Data

A.i. Club Meeting

Tuesday, October 14, 2020

Room 120A

11:00 PM

Special Guest

R.S.V.P. www.towerofhanoi.com

Reasoning With Data

A.I. is about Solving Problems. Puzzle of the Day is for Practice

©

Many, many questions in one small notice



Problem Solving Skills 01

Online - LIVE

61 of 70

01:00

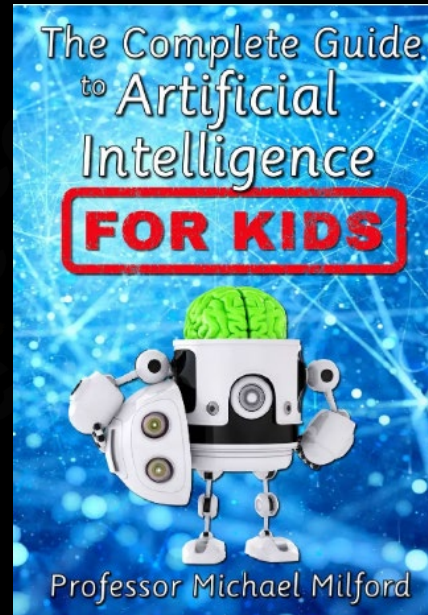
A.I. 1370



PS01
#1 of 16
Online

A.i. is being taught at all levels, everywhere

<https://www.scribd.com/document/400573590/AI-Guide-Ebook-pdf>



This is a better way to move into STEM

©

Where are these kids going to be at your age



A.I. is now a household word/practice

Online - LIVE

62 of 70

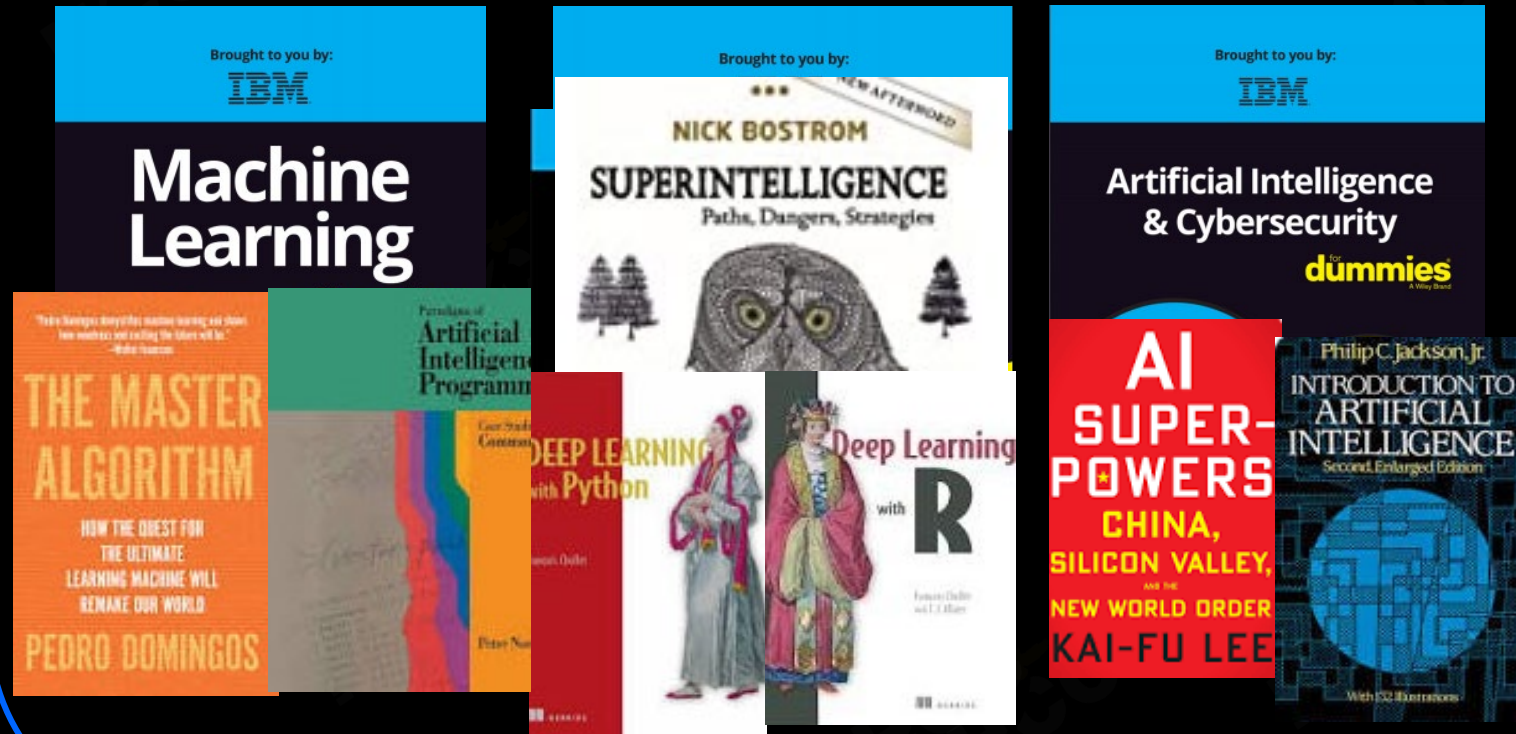
01:00

A.I. 1370



Ref 01
#1 of 16
Online

We have assembled the very “Finest” Materials and SME’s



You will have access to an incredible Reference and Resources Archive

©

The References Section is full of goodies ◀ ▶ References, Resources and Reading

Online - LIVE

63 of 70

01:00

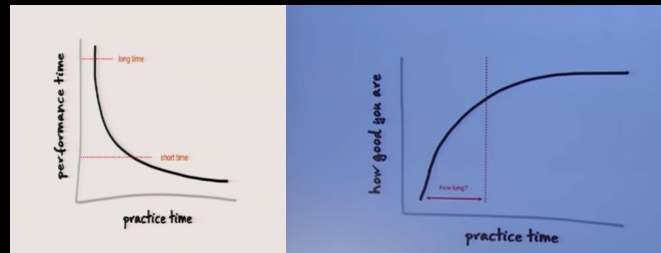
What does It Take To Learn a New Skill?

CSU Ted Talk: <https://www.youtube.com/watch?v=5MgBikgcWnY>

10,000 Hours = fulltime job for 5 years = picking up a new skill studied by K. Anders Erickson, Fla State University – He studied experts primarily sports champions, Chess Champions, people considered to be the “Best” in narrow, competitive fields. What does it take to get to the top? Lots of Expert Level Practice. Malcom Gladwell – “Outliers” It takes 10,000 hours to learn something at an expert level.

If we are going to be effective learners (not super experts) then the subject about learning a new skill is about getting reason-ably good or effectively learning in about 20 hours or 45 min a day for a month. There is a way to practice efficiently. The only barrier has been the perception of Intellectual Superiority or being an expert (Ego). The Real barrier to learning is emotional and not, intellectual nor is it time consuming. Enough practice to be able to self correct errors = right vs wrong things to do.

Josh Kaufman TED Talk



Becoming an expert (10,000 hrs) Becoming an effective learner reverses the curve

©

Get Acquainted Data Game using Manual Ai.

We are going to take some data (basic info), first we are going to divide into teams of 4 (if we have 24 people = 6 teams)

- A. Name your team, Pick a theme song, Team Colors and a Team Motto (Goal Statement)
- B. Next, we are going to collect some data, manually.

1. Name, Picture, Age, e-mail, phone, Hobbies, Status (academic)
2. Data we will falsify – S.S.#, Mother's Maiden Name, Marital,
3. Birthdate, Street Address, Health Issues, Employer, Bank,
4. Salary, Country of Origin, Spouse's Name, Spouse's S.S., children,
5. Type of Car/year, Favorite Music Website, Facebook Page,
6. Last Amazon Order, \$ spent per week eating out, Web Accounts,
7. Type of Phone (Monthly Plan), Provider, Voter, Party, Insurer,

We are going to use a Spreadsheet to collect and hold this information.
Your Job is to build a Table/Column Spreadsheet to contain your team's info

©

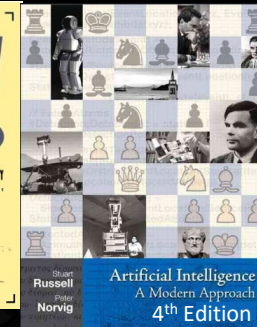
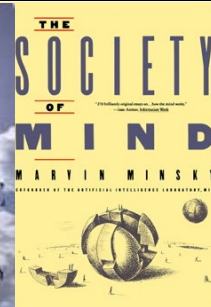
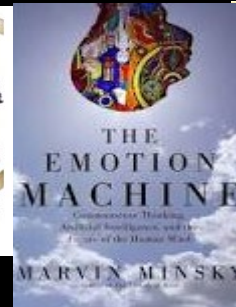
These are Excellent Videos that Paint the Broad Picture of A.I.

1. Patrick Winston (Open Courseware at MIT) <https://www.youtube.com/watch?v=TjZBTDzGeGg>
2. Lex Fridman (MIT) Deep Learning – <https://www.youtube.com/watch?v=OVH1Lim8gL8>
3. Nvidia Drive – Platform <https://www.youtube.com/watch?v=iTZ9GoN2Q0k>
4. IBM – Watson <https://www.youtube.com/watch?v=yXcDir9Y9CI>
5. Peter Bock – Creativity <https://www.youtube.com/watch?v=CpNfy7AUPl4>
6. Explauto – Learning like children learn <https://www.youtube.com/watch?v=NOLAwD4ZTW0>
7. Sophia – Physical Embodiment https://www.youtube.com/watch?v=Bg_tJvCA8zw
8. Sophia - Sophia robot 2020 - <https://www.youtube.com/watch?v=WPhPrnw4QKk>
9. Sophia – Neuroscience <https://www.youtube.com/watch?v=Gmr4i6ZcSdo>
10. Charlie Rose, The Brain Series <https://charlierose.com/collections/3/clip/15731>

This is your homework

©

References, Resources and Reading Material

 Artificial Intelligence Short History, Prese... How Artificial Intelligence is Transformin... TechnicalAgenda Elements of Information Theory 2nd Editi... ibm-machine-learning-for-dummies-ib... ibm-ai-and-cybersecurity-fd-ibm-limite... nvidia-ampere-architecture-whitepaperFridman <https://www.youtube.com/watch?v=O5xeyoRL95U>Fridman <https://www.youtube.com/watch?v=0VH1Lim8gL8>Elon Musk <https://www.youtube.com/watch?v=dEv99vxKjVI>Ian Goodfellow: <http://www.deeplearningbook.org><https://developer.ibm.com/technologies/artificial-intelligence/articles/cc-beginner-guide-machine-learning-ai-cognitive/><https://www.upgrad.com/blog/neural-networks-for-dummies-a-comprehensive-guide/><https://ocw.mit.edu/resources/res-ll-005-mathematics-of-big-data-and-machine-learning-january-iap-2020/class-videos/><https://ocw.mit.edu/resources/res-ll-005-mathematics-of-big-data-and-machine-learning-january-iap-2020/download-resource-materials/>

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Lecture
Notes 05

A.i. Club Meeting

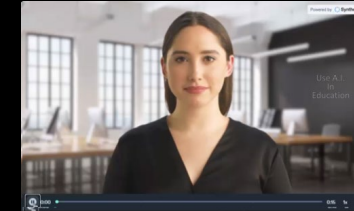
Tuesday, October 14, 2020
Room 120A
11:00 PM
Special Guest

R.S.V.P. www.towerofbabel.com

This is a typical Flyer

Puzzle of the Week
Verify all the
data in this flyer
Make it better

Big idea of the Week
Synthetic Avatar took
30 minutes to make



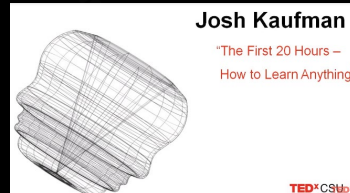
Wednesday, September 23, 2020 6:00 P.M.

MAKE ME LAUGH CHALLENGE

PART 1

Joke of the Week
Wednesday

Key Video
of the Week
The Trinity of A.I.

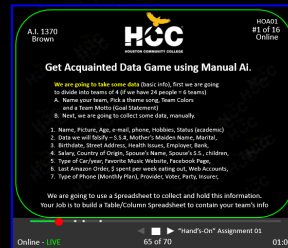


Josh Kaufman

"The First 20 Hours –
How to Learn Anything"

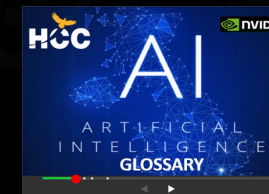
Leadership Skill 01
Learn anything in 20 Hours
Josh Kaufman Ted Talk

Key Article
of The Week
Chihuahua or muffin?

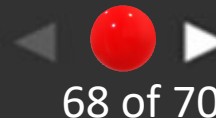


Get Acquainted Data Game
Assignment A01 –
Excel Spreadsheet with info

Lab this Week
A.I. Glossary L01
Term Project



These weekly Gems are the Key
items to focus on for this Week



A.I. GEMS for the Week highlight
the Key Issues

©

A.I. 1370



Challenges
#1 of 16
Online

The Final Exam

We want you to make a Study Portfolio (Throughout the term)

Highlight key ideas

Slide #41



The Slide representing the idea

Deep Learning in Neural Networks: An Overview

Technical Report IDSIA-03-14 / arXiv:1404.7828 v4 [cs.LG] (88 pages, 888 references)

Jürgen Schmidhuber
The Swiss AI Lab IDSIA
Istituto Dalle Molle di Studi sull'Intelligenza Artificiale
University of Lugano & SUPSI
Galleria 2, 6928 Manno-Lugano
Switzerland

8 October 2014

Abstract

In recent years, deep artificial neural networks (including recurrent ones) have won numerous contests in pattern recognition and machine learning. This historical survey compactly summarizes recent work, much of it from the previous millennium. Shallow and deep kernels are distinguished by the depth of their *convolutional* paths, which are chains of possibly non-linear, causal links between actions and effects. To make deep supervised learning also recapitulating the history of backpropagation, unsupervised learning, reinforcement learning & evolutionary computation, and indirect search for short programs encoding deep and large networks.

LATeX source: <https://www.idsia.ch/~juergen/deeplearning2014.tex>
Complete BibTeX file (888 KB): <https://www.idsia.ch/~juergen/deep.bib>

Highlight Text

Preface

This is the preprint of an invited *Deep Learning* (DL) overview. One of its goals is to assign credit to those who contributed to the present state of the art. I acknowledge the limitations of attempting to achieve this goal. The DL research community itself may be viewed as a continually evolving, deep network of scientists who have influenced each other in complex ways. Starting from recent DL results, I tried to trace back the origins of relevant ideas through the past half century and beyond, sometimes using "local search" to follow citations of citations backwards in time. Since not all DL publications properly acknowledge earlier relevant work, additional global search strategies were employed, aided by consulting numerous neural network experts. As a result, the present preprint mostly consists of references. Nevertheless, through an expert selection bias I may have missed important work. A related bias was surely introduced by my special familiarity with the work of my own DL research group in the past quarter-century. For these reasons, this work should be viewed as merely a snapshot of an ongoing credit assignment process. To help improve it, please do not hesitate to send corrections and suggestions to juergen@idsia.ch.

Integrating Knowledge and Reasoning in Image Understanding

Somak Aditya¹, Yeshou Yang² and Chitta Baral²
¹BigData Experience Lab, Adobe Research, India
²CHISEL, Arizona State University, USA
aditya@adobe.com, [@asu.edu](mailto:y.yang.chitta)

Abstract

Deep learning based data-driven approaches have been successfully applied in various image understanding applications ranging from object recognition, semantic segmentation to visual question answering. However, the lack of knowledge integration as well as higher-level reasoning capabilities with the methods still pose a hindrance. In this work, we present a brief survey of a few representative reasoning mechanisms, knowledge integration methods and their corresponding image understanding applications developed by various groups of researchers, approaching the problem from a variety of angles. Furthermore, we discuss open key challenges in integrating external knowledge with neural networks. Taking cues from these efforts, we conclude by discussing potential pathways to improve reasoning capabilities.

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1 Introduction

From the early years of computer vision research, many researchers realized that prior knowledge could help in different tasks ranging from low-level to high-level image understanding. For example, knowledge about the shape of an object can act as a strong prior to segmentation tasks [1], or knowledge about the most probable action given a subject and the object can aid in action recognition tasks [2, 3]. In the recent era of data-driven techniques, most of this knowledge is hoped to be learned from training data. While this is a promising approach, annotated data can be scarce in certain situations, and many domains have a vast amount of knowledge encoded in form of text, structured or unstructured that can be utilized in such cases. Utilization of background knowledge in data-driven situations is one of the manners that accelerate the development of approaches that can utilize such knowledge. Additionally, the lack of reasoning and inference capabilities (such as common sense, causal graphs) of deep learning systems recently started to surface in various ESSEM [15]. Motivated by these challenges, our goal in this paper is to present a survey of recent works (including a list of our work) in image understanding where knowledge and reasoning plays an important role. While discussing these reasoning mechanisms, we introduce corresponding reasoning mechanisms, knowledge sources, and argue the merits behind their choice. Lastly, we discuss different mechanisms that integrate external knowledge sources directly with deep neural networks.

Figure 1: The diagram shows the information hierarchy for images and the knowledge associated with each level of information.

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To understand what knowledge is meaningful in images, we can look at the different types of knowledge that relate to different levels of the semantic hierarchy induced by a natural image. Natural images are composed of a natural image is composed of objects and regions. Each object is composed of parts that could be objects themselves and regions can be composed of semantically meaningful sub-regions. This compositionality of images can be used to define parts for objects (and higher level concepts). We show a diagram representing the semantic hierarchy induced by a natural image in Figure 1. Different types of knowledge might be related to the various levels of the semantic hierarchy and their parts to higher-level semantics (abstract concepts, actions). For example, the knowledge of objects and regions might be applicable in these following levels of semantics: (i) object, region and their attributes; (ii) object-object or object-region interactions, relations and actions; (iii) high-level commonsense knowledge (about events, activities). In this work, we review the sources around different image applications ranging from object classification to question answering. As each new reasoning engine and know-

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